Analysis of Project Green Light and Gentrification in Detroit

Jennifer Gleit

Abstract:

Project Green Light is a unique crime deterrence and surveillance system started in Detroit in 2016. The objective of this study is to analyze the impact of Project Green Light in relation to gentrification in the city. To accomplish this, python scripts were utilized to retrieve demographic, household, and economic data from the American Community Survey through API calls. Using this data and Hirsch and Schinasi's method, census tracts were classified as ineligible for gentrification, not gentrified, gentrifying, or intensely gentrifying for each applicable year. These classifications were then mapped and overlaid with the locations of green lights that existed in that year in ArcGIS Pro. Correlation and regression analyses were then performed in R to compare the relationship between the number of green lights and the gentrification level in a census tract. No meaningful correlation was found, and challenges in normalizing the data prevented successful regression analysis. Rather than using the aggregated classification, the potential to instead perform a regression analysis using the elements involved in that classification, including how rent, home value, and proportion of residents with a college education have changed over time is discussed.

Background:

Project Green Light Detroit is a unique partnership program between the city of Detroit, the Detroit Police Department, and the city's business community, intended to prevent and solve crime. The program is supposed to build upon the common usage of closed-circuit television cameras. Businesses that choose to participate and purchase the necessary equipment have real-time surveillance monitoring at Detroit Police Headquarters and receive higher priority in response to calls. These locations also typically display large signs advertising their participation in the program, as well as literal flashing green lights, intended to serve as a crime deterrent (Circo et. al., 2020). The first green light went live in 2016, and the program has continued to expand ever since. Project Green Light is the first program of its kind, so analysis of its impacts is especially important if other municipalities are considering implementing similar programs.

I hope to complete an exploratory analysis of the impact of the green lights, looking to identify patterns in their implementation and comparing the attributes of nearby areas before and after installation of a significant number of lights, specifically elements that can be used to analyze gentrification, such as housing costs and education levels.

Ultimately, the question I seek to answer is whether installation of green lights in an area is a precursor to gentrification, a consequence of gentrification, or unrelated entirely.

Methodology

Data and Materials

Retrieve and Prepare Data

The first step to begin analysis is to gather and prepare the data. The TIGER/Line shapefile of 2010 census tracts for the state of Michigan is clipped to the city of Detroit using the City of Detroit Boundary shapefile from Detroit's Open Data Portal. Next, using a script written in python and run through the PyScripter program, the Census Data Application Programming Interface (API) is called to retrieve data elements from the American Community Survey (ACS) 5-year data sets and store them in csv files.

In order to use the gentrification classification system designed by Hirsch and Schinasi, the following attributes are needed for each census tract, for each year - number of residents, median household income, median gross rent, median home value, and proportion of residents with a college education. All but the last of these elements is already present within the ACS 5-year data. To calculate the final attribute needed, the proportion of residents with a college education, the following data elements are retrieved – total population age 25 or older, total population age 25 or older with an associate's degree, total population age 25 or older bachelor's degree, total population age 25 or older master's degree, total population age 25 or older professional degree, and total population age 25 or older doctoral degree. The latter five of these elements are summed together to get the population age 25 or older with a college education, then divided by the total population age 25 or older and multiplied by 100 to get the percentage of residents with a college education. The API is called to retrieve data and the proportion of residents with a college education calculation is performed for each year from 2014 through 2019.

After this, again using a python script, the csv files containing the ACS data for each year are converted to geodatabase tables. They are then joined to the TIGER/Line shapefile of census tracts in the city of Detroit on the tract ID and the new features are exported,

resulting in one polygon shapefile per year containing all census tracts in the city of Detroit and all necessary ACS data for that year.

Calculate Differences Between Years

Next, because Hirsch and Schinasi's method of gentrification level classification involves comparing relative increases over time, we need to calculate the differences in each data element between years. Using a python script, for each year from 2015 through 2019, new fields are added to the shapefiles to store the difference in median household income, difference in percentage of residents with a college education, difference in median gross rent, and difference in median home value between that year and the year prior. Each year's shapefile is joined by the tract ID to the prior year's shapefile and for each of the applicable data elements the prior year's value is subtracted from the current year's value to get the increase for each census tract.

Summarize Differences Between Years

The differences in median household income, percentage of residents with a college education, median gross rent, and median home value between each year now need to be aggregated. Using a python script, a new geodatabase table is created to store this summarized information, including the years from and to, the median change in percentage of residents with a college education between the two years, the median, or 50th percentile, increase in home value between the years, the median, or 50th percentile, increase in gross rent between the years, the 75th percentile increase in home value between the years, the 75th percentile increase in gross rent between the years, and finally the 75th percentile of household income for the current year. The resultant summary table can be seen below in Table 1:

Table 1: Aggregated Yearly Summarized Data

Year From	Year To	Median Increase in Percentage of Residents with a College Education	Median Increase in Home Value	75th Percentile Increase in Home Value	Median Increase in Gross Rent	75th Percentile Increase in Gross Rent	75th Percentile of Household Income
2014	2015	0.230173272	-1000	1925	-2	21.25	31306
2015	2016	0.324214166	0	3925	-0.5	32.25	31657
2016	2017	0.220319814	850	5625	13	42	33044.5
2017	2018	0.344448612	1800	7625	18	49	34984.25
2018	2019	0.471871906	2200	8100	16	53.25	36117.25

The summarized values are generated using a python script and the NumPy package. Each feature class is converted to a NumPy array, and the NumPy median and percentile functions are utilized to calculate these aggregate values.

Evaluate Each Census Tract in Comparison to Overall Yearly Change

Once the summary data for each year's change has been created, every census tract's individual change must be compared to the overall yearly change. Using another python script, for each year from 2015 to 2019, for each census tract, the increase in percentage of residents with a college education is compared to the median increase for the year and marked as above below. Similar calculations are done to mark whether the increases in gross rent and home value, respectively, for each census tract are below the median increase for the year, between the 50th and 75th percentile increase for the year, or above the 75th percentile increase for the year. Finally, the median household income for the year is compared with the 75th percentile of household income for the year. The results of each of these comparisons are stored in new fields added to the yearly shapefiles.

Classify Each Census Tract's Level of Gentrification

Finally, using Hirsch and Schinasi's method of classification, which can be seen below in Figure 1, each census tract for each year is classified as Ineligible, Not Gentrified, Gentrification, or Intense Gentrification.

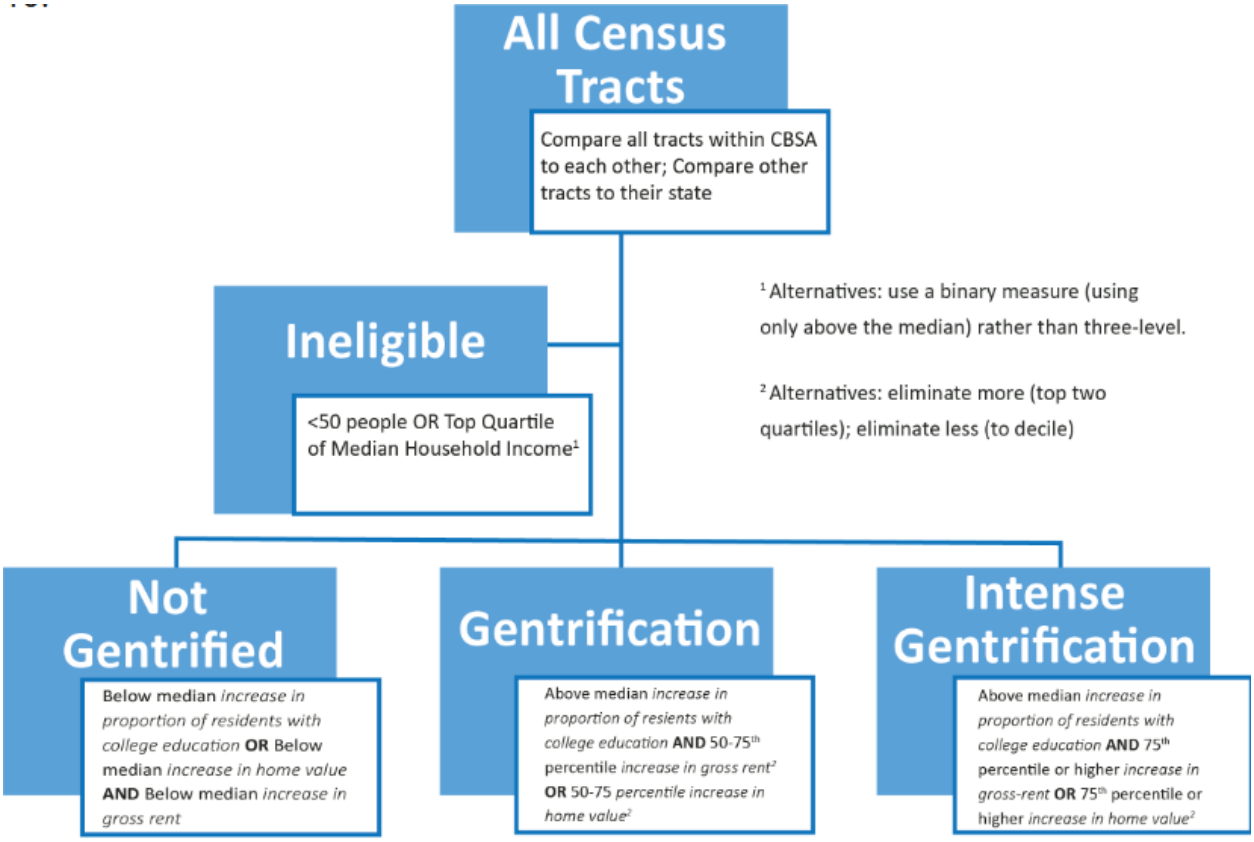


Figure 1: Diagram showing gentrification level classification (Hirsch & Schinasi, 2019)

The classification is done using a python script to evaluate the comparison fields created in the previous step. A new field is created in each yearly shapefile to store the classification, which is represented as an integer. Tracts with fewer than 50 residents or an income above the 75th percentile are classified as Ineligible (0). Tracts with below the median increase in percentage of residents with a college education, or both a below median increase in home value and a below median increase in gross rent are classified as Not Gentrified (1). Tracts with above the median increase in percentage of residents with a college education and either an increase in gross rent between the 50th through 75th percentile or an increase in home value between the 50th and 75th percentile are classified as Gentrification (2). Tracts with above the median increase in percentage of residents with a college education and either a 75th percentile or above increase in gross rent or a 75th percentile or above increase in home value are classified as Intense Gentrification (3).

Combine Classified Census Tracts with Green Light Data

A shapefile of green light locations can be downloaded from the City of Detroit Open Data Portal. This dataset includes the date that each light went live. In ArcGIS Pro, the Export Features tool can be used with a filter to divide out the lights that went live in each year.

These new point feature layers of green lights are used in combination with the classified census tract features to create the maps shown in Figures 5 through 13 in the Additional Tables and Figures section. These figures represent a time series, one for each year 2015 (the year prior to the first green light installation) through 2019, and identify newly added lights for each year as well as color coding each census tract based on its level of gentrification.

Challenges and Limitations

Due to the differences in census tracts between each decennial census, a crosswalk must be completed to compare ACS data using the 2010 census tracts to ACS data using the 2020 census tracts. This was not completed for this project due to time constraints. Instead, the TIGER/Line shapefile of 2020 census tracts for the state of Michigan was downloaded, and all the data preparation steps were repeated for years 2021 through 2024, the most recent year for which ACS data was available. As the data for the year 2020 could not be compared with the prior year, 2019, without the crosswalk, this year was omitted.

Analysis and Methods

The initial goal of this analysis was to perform a spatial regression analysis using the number of green lights installed in a census tract as the predictor variable and the tract's level of gentrification as the response variable. The Summarize Within tool in ArcGIS Pro was used for each year to get the number of green lights installed as of that year that were contained within each census tract. As the original proposal was to use only number of green lights and level of gentrification, not to include time as a variable in the regression analysis, all of the yearly summaries were then merged together in ArcGIS Pro. This table was then exported to Excel, where it was saved in csv format, then loaded into R Studio. As intended in the proposal, below in Figure 2 can be seen a scatter plot created in R displaying the number of green lights and level of gentrification.

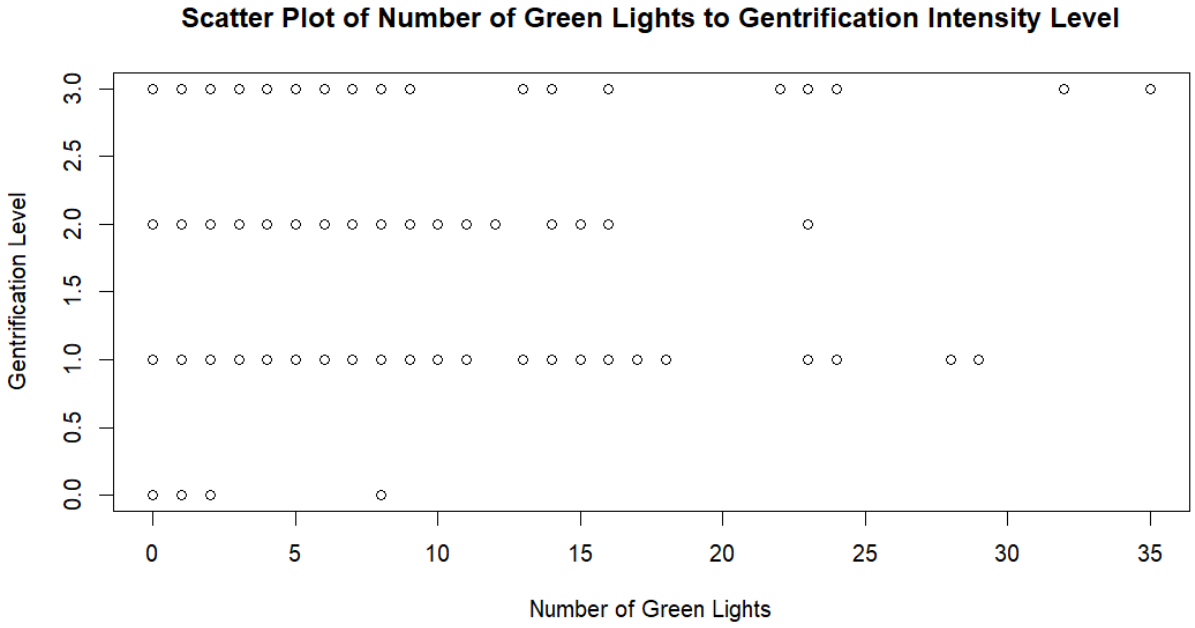


Figure 2: Scatter plot of number of green lights and gentrification level

This scatter plot only displays one point for each set of number of green lights and gentrification level that occurs in the data, but does not show the relative frequencies, or how many census tracts had that number of green lights with that gentrification level. Unfortunately, due both to this limitation and the nominal categorization of gentrification level, this scatter plot does not yield any meaningful insights about the data.

At this point, the regression analysis began. The first step was to evaluate the assumptions, beginning with that the data must be normally distributed. This proved to be a difficult assumption to meet. Using a combination of both Excel functions and R functions, a variety of different data transformations were tested on the full data set containing all years, including log base 10, natural log, square root, cube root, and squared data. For each data transformation, a histogram was generated in R to visually evaluate for normality, and the Shapiro-Wilk test was also performed in R. None resulted in p-values less than any reasonable alpha value.

As can be seen below in Figure 3, the number of green lights is heavily positively skewed, with the majority of census tracts having 0 or 1 green light. Therefore, these tracts were removed, and various data transformations were tested again. Even after removing tracts with 0 or 1 light, the heavily positive skew remained, as can be seen in Figure 4. Removing tracts with two lights was also tested, as well as using the interquartile range method to remove other outliers.

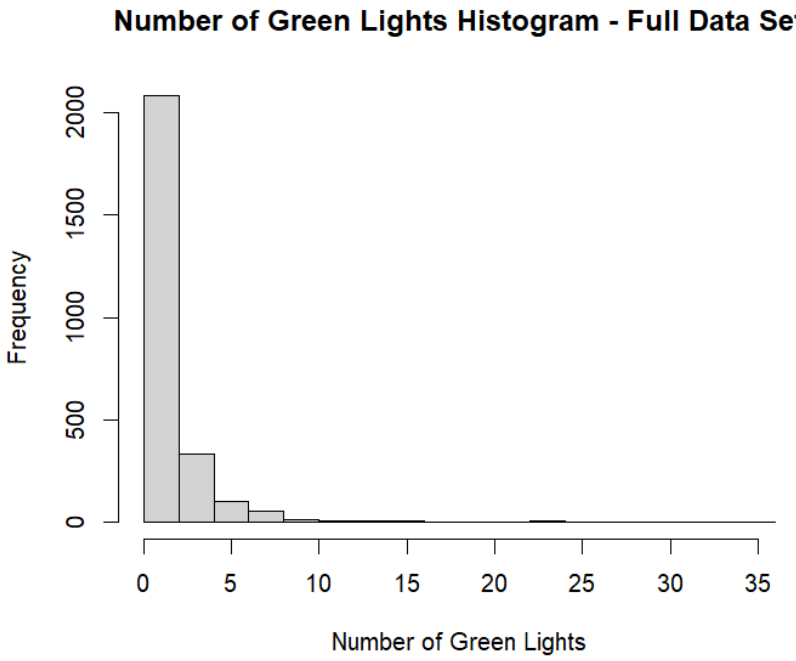


Figure 3: Histogram showing frequency of each number of green lights

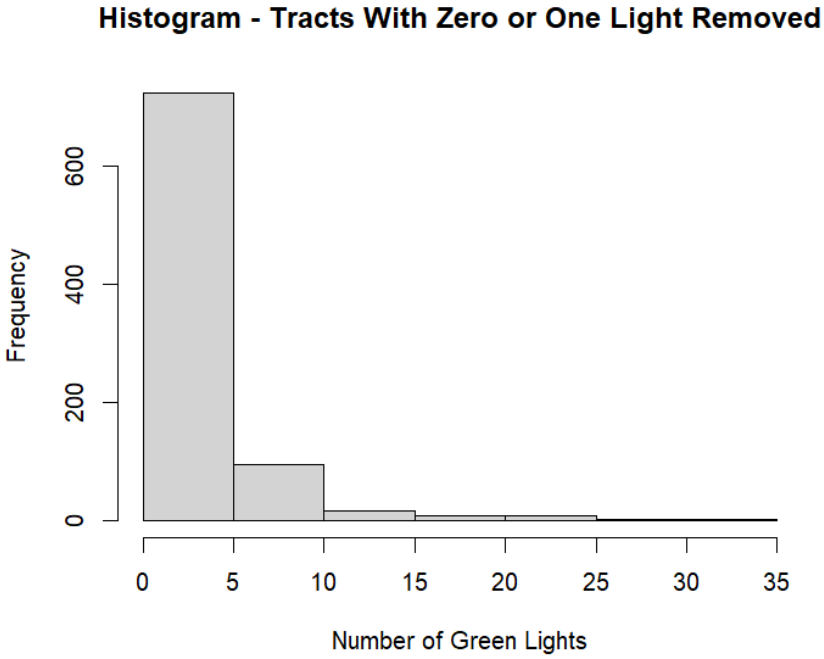


Figure 4: Histogram showing frequency of each number of green lights, excluding tracts with 0 lights or 1 light

Thinking that perhaps evaluating only one year at a time would result in more normally distributed data, only the 2019 data and only 2024 data were each tested, both times again testing removing tracts with 0 or 1 light and various data transformations. Again, none resulted in reasonable p values for the Shapiro-Wilk test. A table of some of these results can be found in Table 2 in the Additional Tables and Figures section.

Results

Overall, the findings show that while the gentrification level classification system utilized is useful to create maps that can be used to visually identify spatial patterns, it is not a meaningful variable for use in regression analysis. No significant correlation was found between the number of green lights and the gentrification level. Specifically, using R to calculate the correlation coefficients between the number of green lights and the level of gentrification, a coefficient of 0.03632753 is calculated using the Pearson method, and a coefficient of 0.02349261is calculated using the Spearman method. These small positive values close to zero indicate only a weak positive correlation between number of green lights and this form of gentrification classification.

However, this does not necessarily mean a relationship between gentrification and green lights does not exist. It is possible using only the overall gentrification classification is too high level and simplistic to be meaningful. Therefore, a potential next analysis to perform would be to utilize the variables that went into the calculation of the overall gentrification level, rather than simply the category itself. To do this, the predictor and response variables would need to be flipped, so that the median increase in rent, median increase in home value, and median increase in proportion of residents with a college education would be predictor variables, and the number of green lights the response variable.

Reflection

In the future I would certainly leave even more time than the additional time I had already allotted for data gathering and transformation and would also develop a more clear-cut naming convention for myself. I had many assorted files stored in different folders throughout this project, so in the future I hope to be more organized in my file system. I also hope to be more organized in my testing documentation, and to take more notes throughout the process, so that I will not need to repeat tests.

Overall, while I was unable to complete the analysis I had originally proposed, I feel that I still learned a lot while completing this project. I struggled a lot with getting data to fit to normal distribution, but I think I gained a much better understanding of data

transformations in the process. I think a lot more can be explored with these data sets, as mentioned in the results section.

I started working on performing this new regression analysis, using median increase in rent, median increase in home value, and median increase in proportion of residents with a college education as predictor variables and the number of green lights the response variable. Using my increased knowledge of data transformations, I was able to transform the median increase in rent to pass the Shapiro-Wilk test with a reasonable p-value.

As some of the rent data had negative values, I started with adding (minimum increase in rent * -1) + 1 to each value to shift all values to be positive. Then, I performed a log transformation on the data and used the interquartile range method to remove outliers. This resulted in a normally distributed data set.

Next, I hope to perform similar transformations to the increase in home value and increase in proportion of residents with college education. I will need to check for correlation between all variables and pay special attention to the increase in home value and the increase in rent, as logically these seem likely to be highly correlated. After this, I hope to be able to fit a multi-variate linear regression model using these variables to predict the number of green lights in a census tract.

Additional Tables and Figures

Table 2: Results of normality tests for different subsets and transformations of the number of green lights per census tract

Data Subset	**Data Transformation**	**W**	**p-value**
Full data set	No data transformation	0.55308	< 2.2e-16
Full data set	Log transformation	NaN	NA (not possible to take log of 0)
Full data set	Square root transformation	0.81721	< 2.2e-16
Tracts with 0 or 1 light removed	No data transformation	0.55917	< 2.2e-16
Tracts with 0 or 1 light removed	Log transformation	0.83845	< 2.2e-16
Tracts with 0 or 1 light removed	Square root transformation	0.72702	< 2.2e-16
Only data for year 2019	No data transformation	0.54225	< 2.2e-16

Only data for year 2019	Square root transformation	0.82613	< 2.2e-16
Only data for year 2019, tracts with 0 or 1 light removed	No data transformation	0.50022	< 2.2e-16
Only data for year 2019, tracts with 0 or 1 light removed	Log transformation	0.77225	5.966e-12
Only data for year 2019, tracts with 0 or 1 light removed	Square root transformation	0.65544	6.511e-15
Only data for year 2024, tracts with 0, 1, or 2 lights removed	No data transformation	0.61992	< 2.2e-16
Only data for year 2024, tracts with 0, 1, or 2 lights removed	Log transformation	0.86216	1.326e-09
Only data for year 2024, tracts with 0, 1, or 2 lights removed	Square root transformation	0.76051	3.14e-13
Only data for year 2024, tracts with 0 or 1 light removed	Cube root transformation	0.83946	8.386e-13

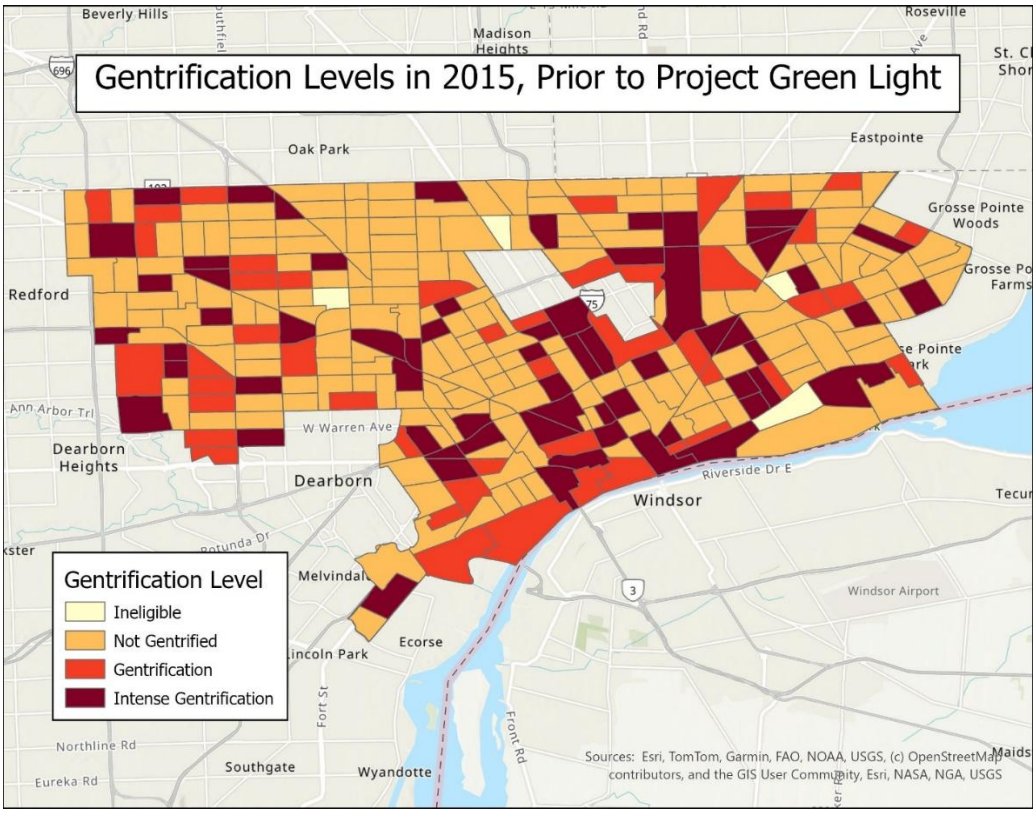


Figure 5: Census tracts in the city of Detroit by gentrification level in 2015

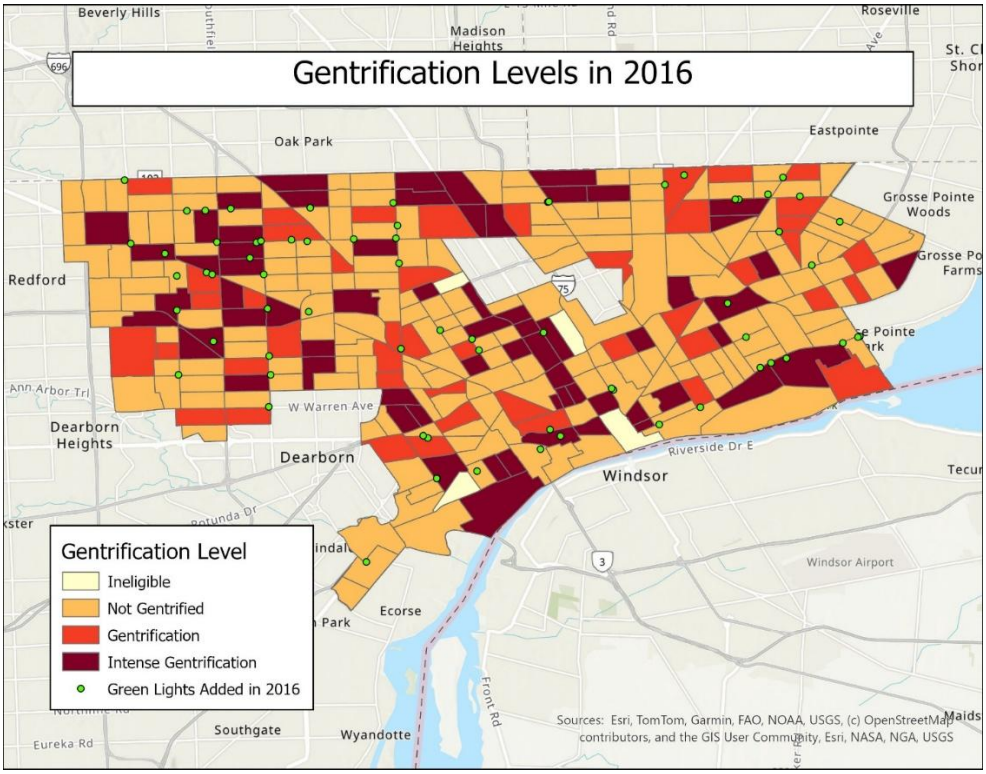


Figure 6: Census tracts in the city of Detroit by gentrification level in 2016, overlaid with locations of green lights in 2016

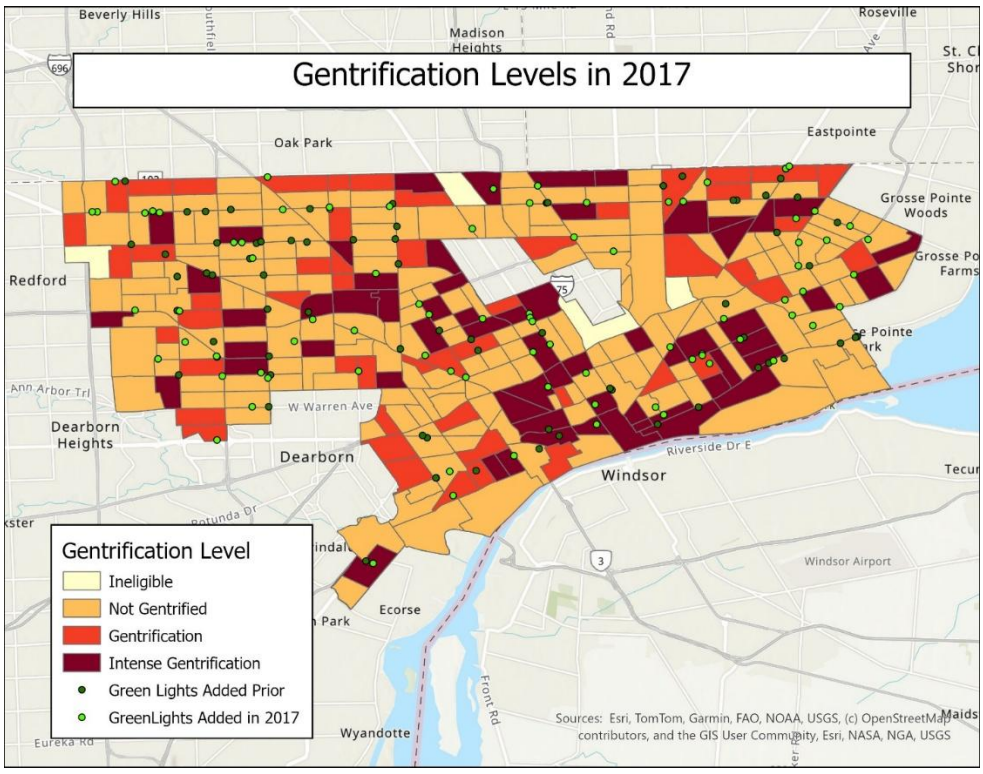


Figure 7: Census tracts in the city of Detroit by gentrification level in 2017, overlaid with locations of green lights in 2017

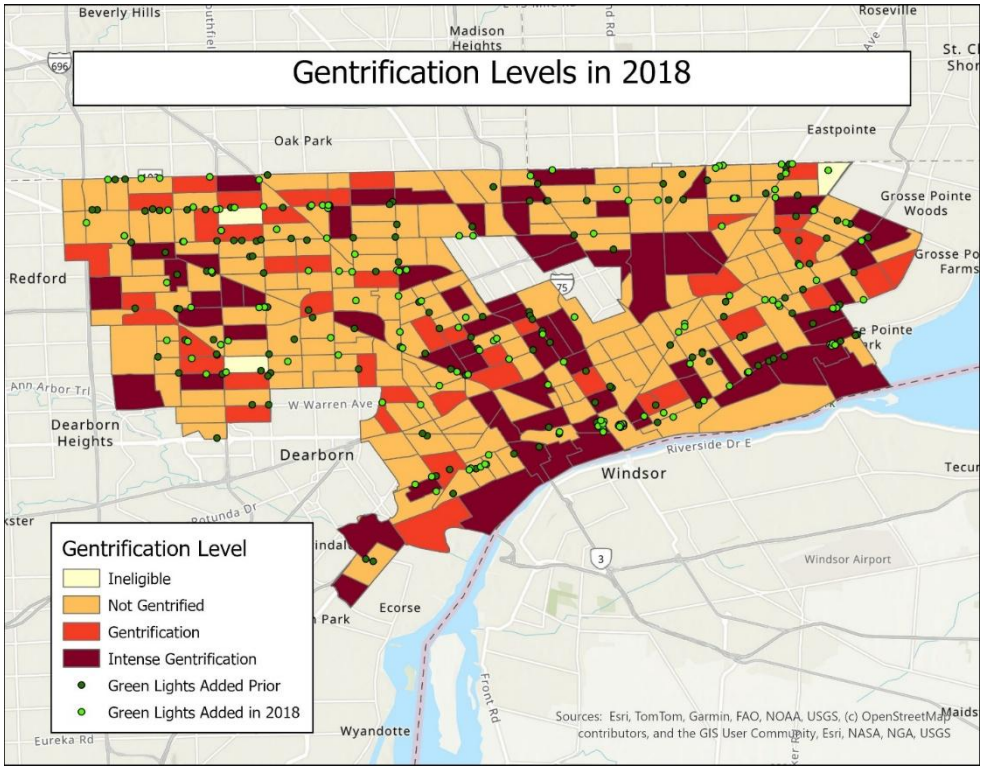


Figure 8: Census tracts in the city of Detroit by gentrification level in 2018, overlaid with locations of green lights in 2018

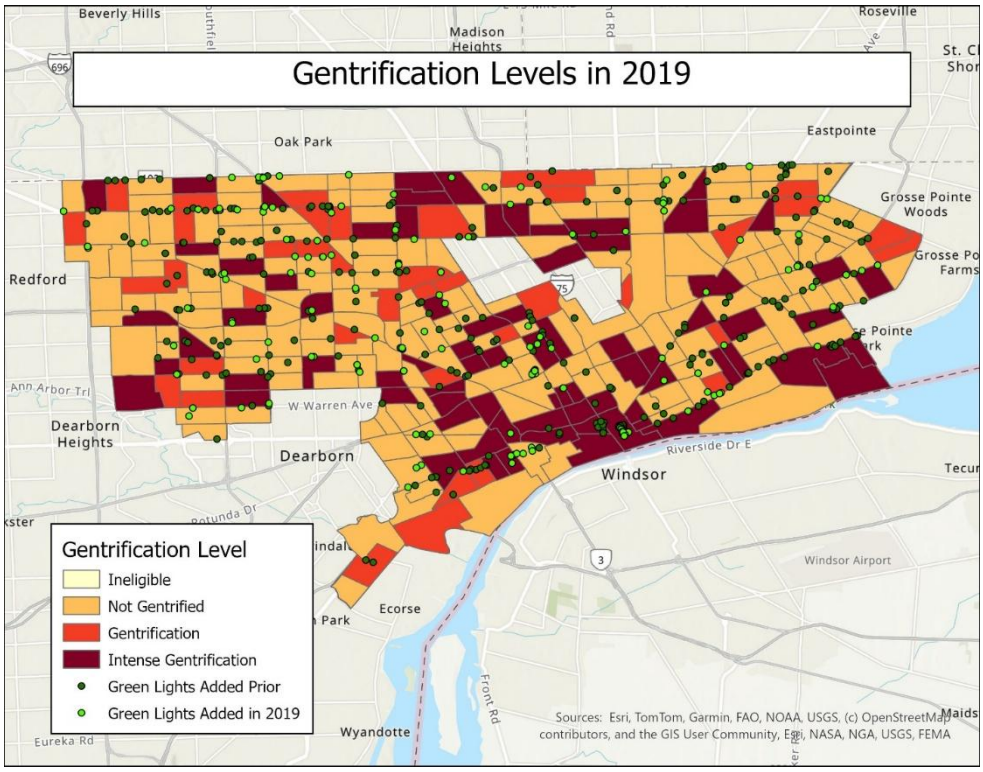


Figure 9: Census tracts in the city of Detroit by gentrification level in 2019, overlaid with locations of green lights in 2019

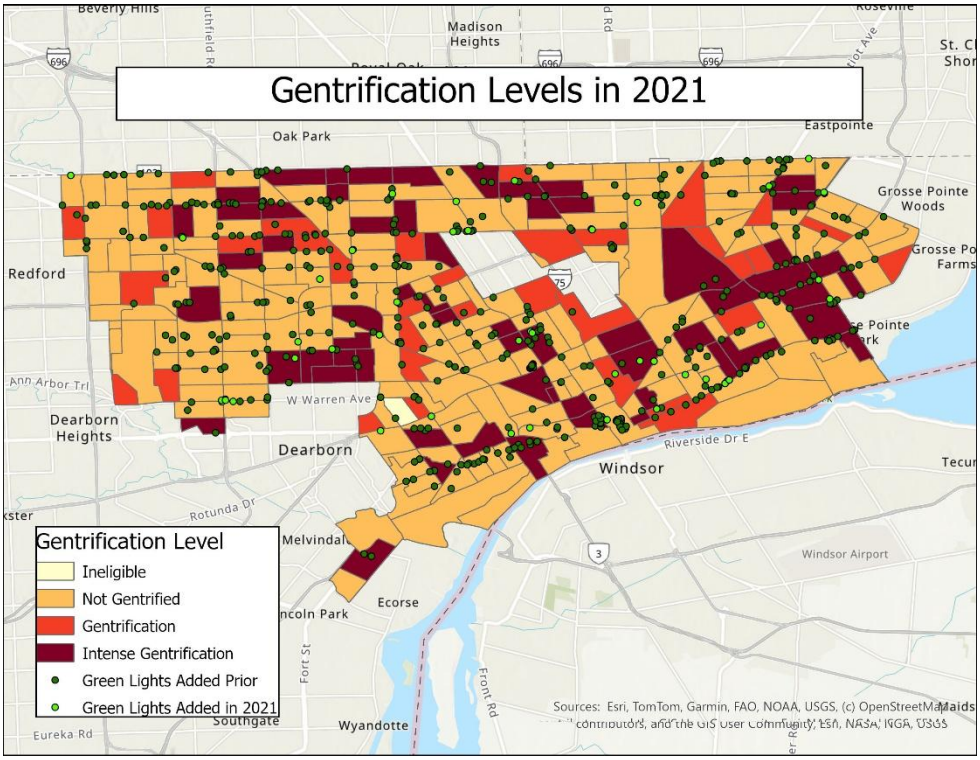


Figure 10: Census tracts in the city of Detroit by gentrification level in 2021, overlaid with locations of green lights in 2021

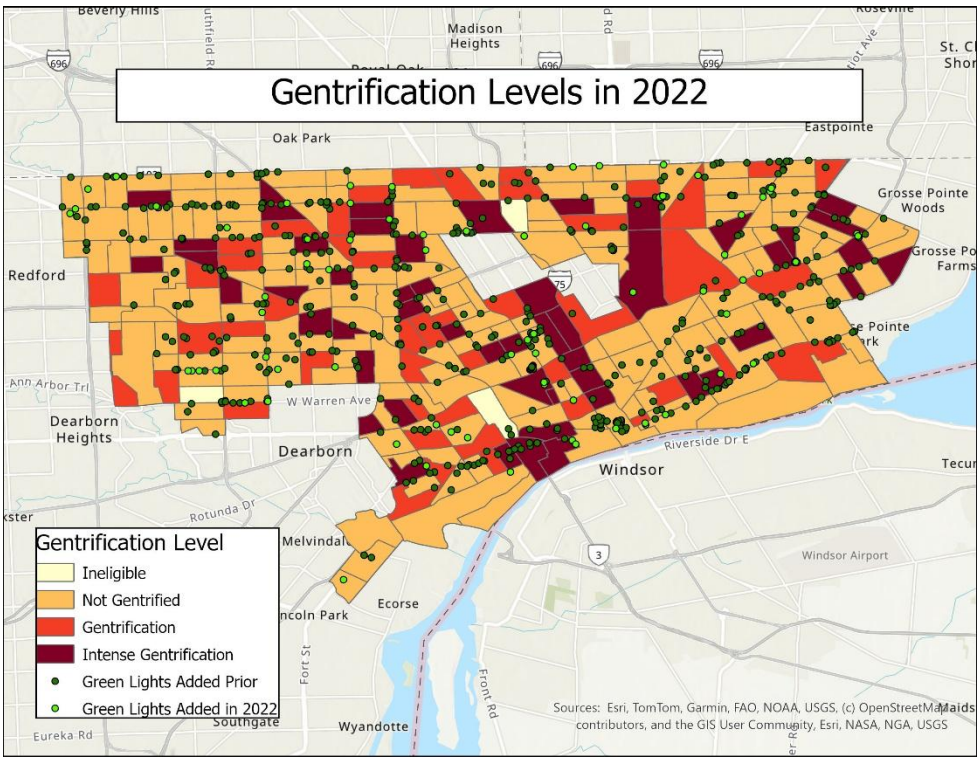


Figure 11: Census tracts in the city of Detroit by gentrification level in 2022, overlaid with locations of green lights in 2022

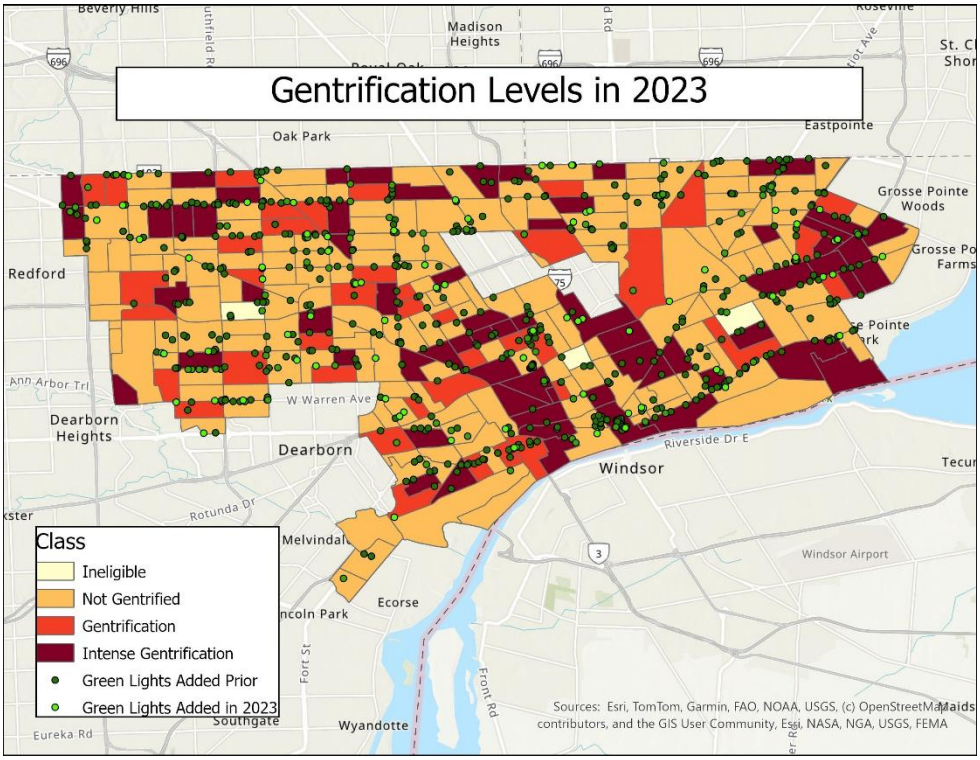


Figure 12: Census tracts in the city of Detroit by gentrification level in 2023, overlaid with locations of green lights in 2023

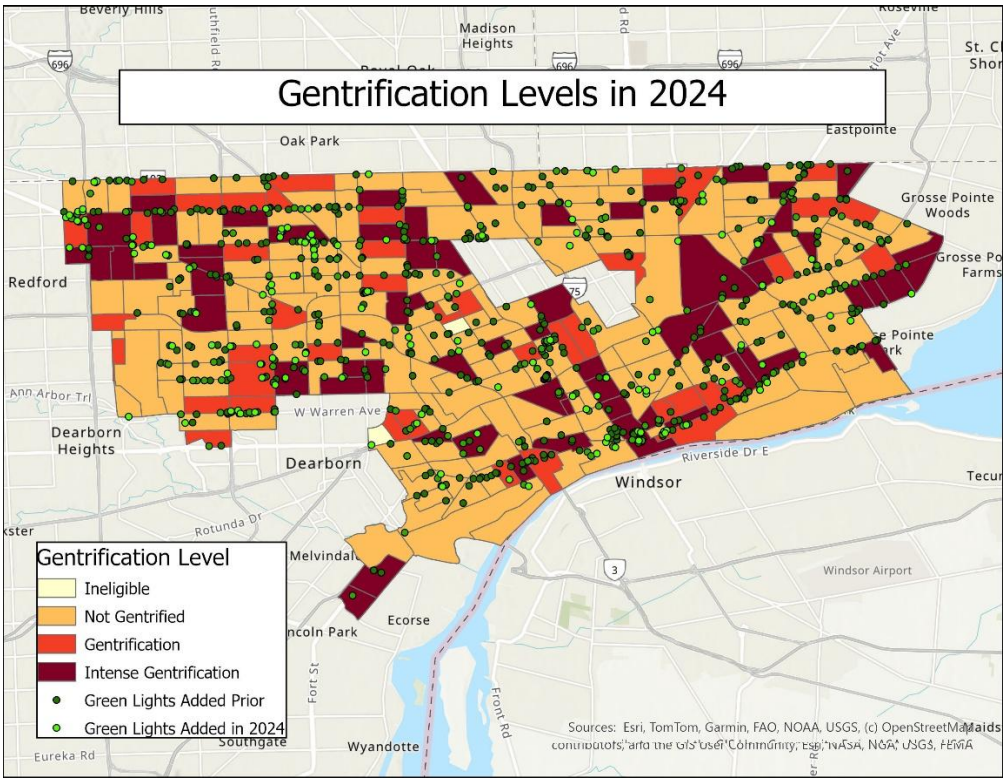


Figure 13: Census tracts in the city of Detroit by gentrification level in 2024, overlaid with locations of green lights in 2024

References

Circo, Giovanni, June Werdlow Rogers, Edmund F. McGarrell, Julie M. Krupa, Alaina De Biasi, Juli Liebler, Shannon Cartwright, and Travis Carter. 2020. *Project Greenlight Detroit: Evaluation Report*. East Lansing, Mich.: Michigan State University.

City of Detroit (May 7, 2026). "Project Green Light Locations" [feature layer]. Scale Not Given. City of Detroit Open Data Portal. Retrieved July 17, 2026 from https://data.detroitmi.gov/datasets/detroitmi::project-green-light-locations/about

City of Detroit (November 12, 2021). "City of Detroit Boundary" [feature layer]. Scale Not Given. City of Detroit Open Data Portal. Retrieved July 17, 2026 from https://data.detroitmi.gov/datasets/detroitmi::city-of-detroit-boundary/about

Hirsch J A, Schinasi L H. A measure of gentrification for use in longitudinal public health studies based in the United States. Philadelphia, PA: Drexel University, Urban Health Collaborative, August 2019

U.S. Census Bureau, "American Community Survey 5-Year Estimates: Comparison Profiles 5-Year," 2014-2024, <http://api.census.gov/data>, accessed on July 19, 2026

U.S. Census Bureau, "ACS_2020_5YR_TRACT_26.gdb", TIGER/Line Shapefiles, 2020, https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-data.2020.html, accessed on July 19, 2026

U.S. Census Bureau, "2010_ACS_5YR_TRACT_26.gdb", TIGER/Line Shapefiles, 2010, https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-data.2010.html#list-tab-1656998034, accessed on July 19, 2026